CE2003 Tut 4

Q1 Complete the following 8-bit signed sums, showing your working. State the conditions that can be detected to determine if a signed overflow has occurred.

23 + 46

-12 + 90

37 - 63

-32 - 62

Q2 You are required to represent the following 6 numbers using the same 10-bit two's complement fixed point format, using as many fractional bits as possible. Determine the representation for each number, and calculate the absolute error for each.

4.3421

-6.115

0.7721

1.9793

-2.44

0.0391

Q3 For the following multiplication operations, determine the wordlength and binary point position of the output, then state the range of values that can be represented. For this question, 3.7 represents a binary number with 3 integer bits and 7 fractional bits.

unsigned 4.6 by 4.6

unsigned 3.9 by 1.11

signed 4.12 by 8.8

signed 1.15 by 1.15

Q4 Determine how many fractional bits you need to represent a number down to a decimal accuracy of 0.0001.

Q5 An arithmetic circuit is described by the Verilog code fragment below:

```
always @ (posedge clk)
    p = (a + b) * c;
```

where a, b and c are 16-bit signed fixed point numbers.

(a) If the most negative and positive values of a and b are -4.381 and +3.965, how many integer and fractional bits should be used for a and b?

(b) If c is always between -0.99 and +0.99, how many integer and fractional bits should be used for c?

(c) Hence, how many integer and fractional bits are there in the result p?